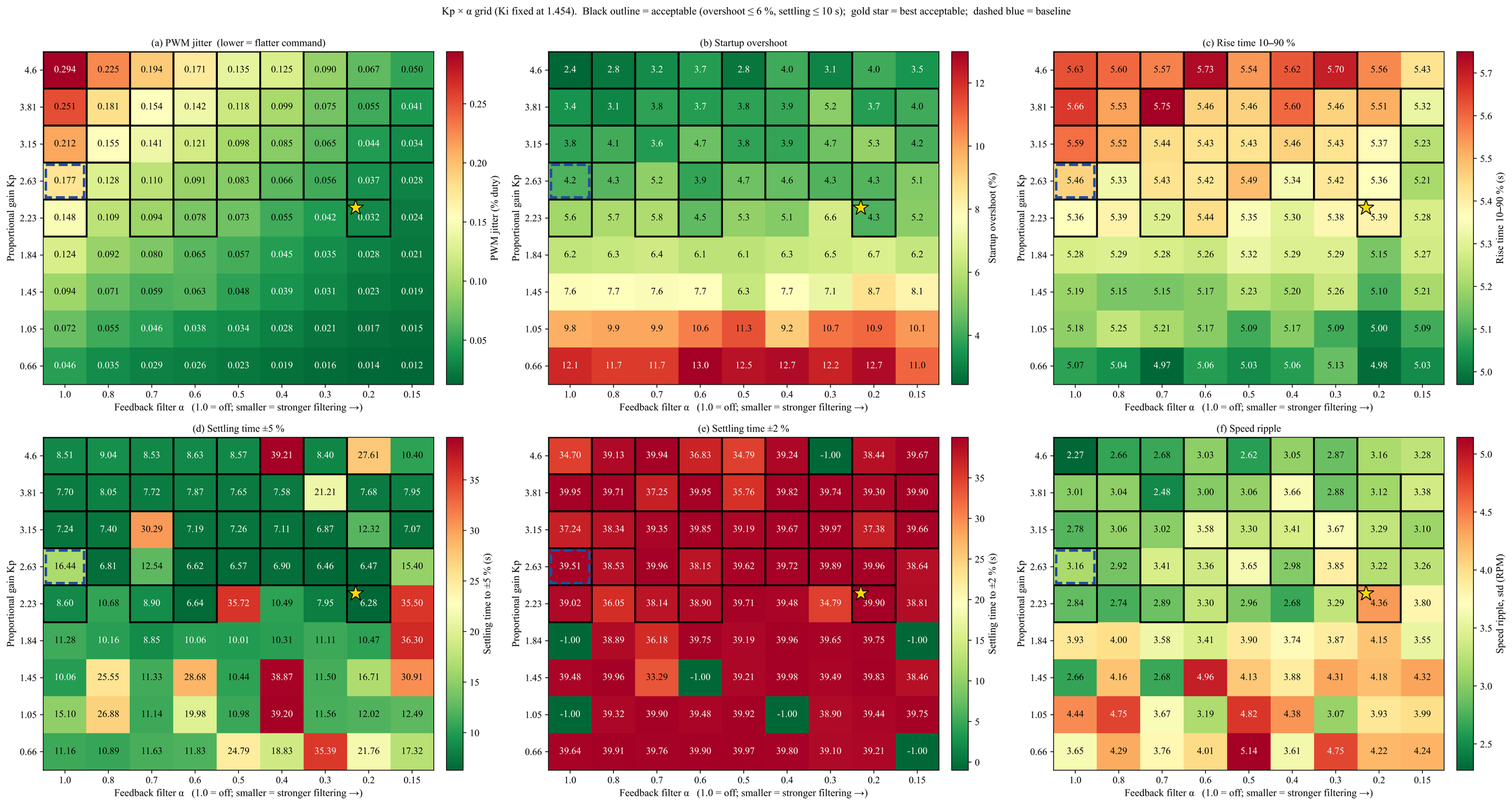

Kp × α grid (Ki fixed at 1.454). Black outline = acceptable (overshoot ≤ 6 %, settling ≤ 10 s); gold star = best acceptable; dashed blue = baseline
(a) PWM jitter (lower = flatter command)
(b) Startup overshoot
(c) Rise time 10–90 %
(d) Settling time ±5 %
(e) Settling time ±2 %
(f) Speed ripple
Proportional gain Kp
Feedback filter α (1.0 = off; smaller = stronger filtering →)
PWM jitter (% duty)
Startup overshoot (%)
Rise time 10–90 % (s)
Settling time to ±5 % (s)
Settling time to ±2 % (s)
Speed ripple, std (RPM)